# Smart Intravenous Dripper

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### Problem Statement

Intravenous drippers are used to inject saline fluids, directly into the veins of the patient. They have to be replaced manually before it completely drains out, or else blood will flow in reverse direction and sometimes air embolism (air-blockage) may even cause a fatal accident.

### Solution

The goal is to prevent such accident by automating this process by using a liquid-level sensor to monitor the liquid level and if the Intravenous dripper runs out of the saline fluid, then the cam actuated by the servo, will stop the flow immediately, thus preventing any such accident to occur and will notify the nurse for further replacement of the intravenous drip.

### How we did

We prototyped our ideation by using majorly texas instruments components.

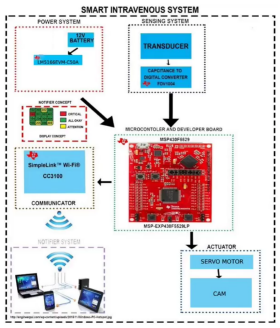


Figure 1: SID architecture

This video shows the working and feedback from **Medical doctors**. We participated in the Department of Science & Technology, Govt. of India & Texas Instruments' **India Innovation Challenge** Design Contest (2016) anchored

by Indian Institute of Managment, Bangalore and out of 11,000+ applicants, we were top 10 teams to reach finals.

### Next What

Being a finalist of IICDC 2016, We received funding for our startup Rekindle Automation Pvt. Ltd. and is currently incubated at NSRCEL, IIM-Banglore. The prototype developed initially for proof of concepts was not reliable, thus now the focus is to develop industrial grade, reliable product.

### The Team

Rohith J.K, Radhakrishnan Jothiram, Wajjahath Habibullah.A and myself..!

Texas Instruments

Certificate of Participation

This is to certify that Rohit JK, Ajay.G, Radhakrishnan.JPS, Wajjahath.H from B .S Abdur Rahman University were Finalist who submitted a proposal entitled "Smart Intravenous Dripper" to DST & Texas Instruments India Innovation Challenge Design Contest 2016, Anchored by IIM, Bangalore.

DST & Texas Instruments Inc.
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Sanjay Srivastava
Director - University Program